

Paper Review

On the power of Chatterjee's rank correlation

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· Shi, H., Drton, M., & Han, F. (2022).

On the power of Chatterjee's rank correlation.

Biometrika, 109(2), 317-333.

Main contributions and outline

Main contributions

- Provide an **alternative proof** of Theorem 2.1 in Chatterjee, S. (2021) that gives an asymptotic representation.
- Show that the test based on Chatterjee's rank correlation coefficient is in fact **rate suboptimal**.
- Show that the tests based on the more established coefficients (D, R, τ^*) are all **rate optimal**.

- Review of Chatterjee's rank correlation
- Other rank correlations considered and their computation
- Local power analysis
- Simulation results
- Conclusion

Review of Chatterjee's rank correlation

Definition of Chatterjee's rank correlation

Definition 1. (Chatterjee's rank correlation)

Let (X, Y) be a pair of random variables, and $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. pairs with the same law as (X, Y) where $n \geq 2$. Rearrange the data as $(X_{(1)}, Y_{(1)}), \dots, (X_{(n)}, Y_{(n)})$ such that $X_{(1)} \leq \dots \leq X_{(n)}$. If there exist ties, choose an increasing rearrangement by breaking ties uniformly at random. Now define

- $r_i = \sum_{j=1}^n I(Y_{(j)} \leq Y_{(i)})$, which assigns the “largest” number to the largest value, and
- $\ell_i = \sum_{j=1}^n I(Y_{(j)} \geq Y_{(i)})$, which assigns the “smallest” number to the largest value.

Let

$$\xi_n(X, Y) := 1 - \frac{n \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{2 \sum_{i=1}^n \ell_i (n - \ell_i)}. \quad (1)$$

When there are no ties, (1) reduces to $\xi_n(X, Y) = 1 - \frac{3 \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{n^2 - 1}$ since the denominator is equal to $n(n^2 - 1)/3$.

- **There is no guarantee that ξ_n always lies within $[0, 1]$.**
- **ξ_n requires a large sample size to achieve reliability.** For instance, let (X, Y) be a pair of continuous random variables such that $Y = f(X)$ with some increasing function f . In this case, $\xi_n = 1 - 3/(n + 1)$.

Theorem 1. (The limit of Chatterjee's correlation coefficient)

If Y is not almost surely a constant, then as $n \rightarrow \infty$, $\xi_n(X, Y)$ **converges almost surely to the deterministic limit**

$$\xi(X, Y) := \frac{\int \text{Var}(\mathbb{E}(I(Y \geq t) \mid X)) d\mu(t)}{\int \text{Var}(I(Y \geq t)) d\mu(t)},$$

where μ is the law of Y . **This limit belongs to the interval $[0, 1]$. It is 0 if and only if X and Y are independent, and it is 1 if and only if there is a measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $Y = f(X)$ almost surely.**

- ξ and ξ_n are not symmetric.
- It also measures a **degree of regression dependence**.
 - ▷ Consider two pairs of **continuous** random variables (X, Y) and $(X, \tilde{Y})$. If $(X, Y) \leq_{\text{disp}} (X, \tilde{Y})$, then $\xi(X, Y) \leq \xi(X, \tilde{Y})$, where $\leq_{\text{disp}}$ denotes the dispersive ordering (Dette et al., 2013).

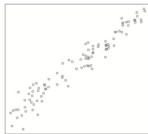
Dispersive ordering

- For the intuition, we only consider continuous random variables.
- We say $F_X \leq_{\text{disp}} F_Y$ if $F_X^{-1}(b) - F_X^{-1}(a) \leq F_Y^{-1}(b) - F_Y^{-1}(a)$ for all $0 < a \leq b < 1$.
Intuitively, **it compares the variability of F_X and F_Y .**
- We denote $(X, Y) \leq_{\text{disp}} (X, \tilde{Y})$ if $F_{Y|X=x} \leq_{\text{disp}} F_{\tilde{Y}|X=x}$ for almost all $x \in \mathcal{X}$. It can be interpreted as “ **Y is more predictable from X than $\tilde{Y}$ is from X** ”.
 - For instance, consider the extreme case that $Y = f(X)$, which gives $F_{Y|X=x}(\cdot) = I(\cdot \geq f(x))$.
 - This implies that $F_{Y|X=x}^{-1}(c) = f(x)$ for all $0 < c < 1$.
 - Thus, $F_{Y|X=x}^{-1}(b) - F_{Y|X=x}^{-1}(a) = f(x) - f(x) = 0$ for all $0 < a \leq b < 1$.
- Recall that $\xi(X, Y) \leq \xi(X, \tilde{Y})$ if $(X, Y) \leq_{\text{disp}} (X, \tilde{Y})$.
- Therefore, we may conclude that ξ **tends to assign higher score to pairs with stronger dependency.**

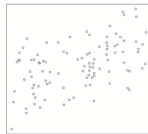
Simulation results for ξ_n



(a) $\xi_n = 0.970$.



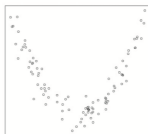
(b) $\xi_n = 0.732$.



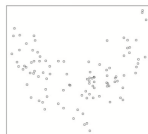
(c) $\xi_n = 0.145$.



(d) $\xi_n = 0.941$.



(e) $\xi_n = 0.684$.



(f) $\xi_n = 0.265$.



(g) $\xi_n = 0.885$.



(h) $\xi_n = 0.650$.



(i) $\xi_n = 0.281$.

Main results for Chatterjee, S. (2021)

Theorem 2. (The limit distribution of Chatterjee's correlation coefficient)

Suppose that X and Y are **independent**. Then $\sqrt{n}\xi_n(X, Y)$ converges to $N(0, \tau^2)$ in distribution as $n \rightarrow \infty$, where τ is a constant that may depend on the law of Y . The number τ^2 is strictly positive if Y is not a constant, and equals $2/5$ if Y is continuous.

- τ^2 can be estimated with a computational cost $O(n \log n)$, which converges to τ^2 almost surely as $n \rightarrow \infty$.
- **There is no asymptotic theory for $\xi_n(X, Y)$ when X and Y are dependent.**
- Consider the hypothesis test $H_0 : \xi = 0$ vs. $H_1 : \xi \neq 0$ when Y is continuous.
 - Given the confidence level α , it is natural to reject the null hypothesis when $\sqrt{n}|\xi_n| > \sqrt{2/5}z_{1-\alpha/2}$.

The limit distribution of Chatterjee's correlation coefficient

- Suppose that Y is **continuous**.
- Recall the formula $\xi_n(X, Y) = 1 - \frac{3 \sum_{i=1}^{n-1} |r_{i+1} - r_i|}{n^2 - 1}$. If X and Y are independent, $(r_1, \dots, r_n)$ is **uniformly distributed over the $n!$ permutations of the integers $1, \dots, n$** .
- It is known that

$$\frac{\sum_{i=1}^{n-1} |r_{i+1} - r_i| - n(n-1)/3}{n^{3/2}} \rightarrow N(0, 2/45).$$

- Simple calculation yields that $\sqrt{n}\xi_n(X, Y) \rightarrow N(0, 2/5)$.
- **The key observation is that $\xi_n(X, Y)$ is just a result of random permutation.**

Other rank correlations considered and their computation

Hoeffding's correlation

Definition 2. (Hoeffding's correlation ; D, D_n)

Hoeffding's correlation measure is defined as

$$D(X, Y) = \int \left\{ F(x, y) - F_X(x)F_Y(y) \right\}^2 dF(x, y).$$

It is unbiasedly estimated by the correlation coefficient

$$\begin{aligned} D_n(X, Y) &= \frac{1}{n(n-1) \cdots (n-4)} \\ &\times \sum_{i_1 \neq \dots \neq i_5} \frac{1}{4} \left[\left\{ I(X_{i_1} \leq X_{i_5}) - I(X_{i_2} \leq X_{i_5}) \right\} \left\{ I(X_{i_3} \leq X_{i_5}) - I(X_{i_4} \leq X_{i_5}) \right\} \right] \\ &\times \left[\left\{ I(Y_{i_2} \leq Y_{i_5}) - I(Y_{i_1} \leq Y_{i_5}) \right\} \left\{ I(Y_{i_3} \leq Y_{i_5}) - I(Y_{i_4} \leq Y_{i_5}) \right\} \right], \end{aligned}$$

which is a rank-based U -statistic of order 5.

Blum-Kiefer-Rosenblatt's correlation

Definition 3. (Blum-Kiefer-Rosenblatt's correlation ; R, R_n)

Blum-Kiefer-Rosenblatt's correlation measure is defined as

$$R(X, Y) = \int \left\{ F(x, y) - F_X(x)F_Y(y) \right\}^2 dF_X(x)F_Y(y).$$

It is unbiasedly estimated by the correlation coefficient

$$\begin{aligned} R_n(X, Y) = & \frac{1}{n(n-1) \cdots (n-5)} \\ & \times \sum_{i_1 \neq \dots \neq i_6} \frac{1}{4} \left[\left\{ I(X_{i_1} \leq X_{i_5}) - I(X_{i_2} \leq X_{i_5}) \right\} \left\{ I(X_{i_3} \leq X_{i_5}) - I(X_{i_4} \leq X_{i_5}) \right\} \right] \\ & \times \left[\left\{ I(Y_{i_1} \leq Y_{i_6}) - I(Y_{i_2} \leq Y_{i_6}) \right\} \left\{ I(Y_{i_3} \leq Y_{i_6}) - I(Y_{i_4} \leq Y_{i_6}) \right\} \right], \end{aligned}$$

which is a rank-based U -statistic of order 6.

Bergsma-Dassios-Yanagimoto's correlation

Definition 4. (Bergsma-Dassios-Yanagimoto's correlation ; τ^* , τ_n^*)

Write $I((x_1, x_2) < (x_3, x_4)) = I(\max\{x_1, x_2\} < \min\{x_3, x_4\})$. Bergsma-Dassios-Yanagimoto's correlation measure is defined as

$$\begin{aligned}\tau^* = & 4P\left((X_1, X_3) < (X_2, X_4), (Y_1, Y_3) < (Y_2, Y_4)\right) \\ & + 4P\left((X_1, X_3) < (X_2, X_4), (Y_1, Y_3) > (Y_2, Y_4)\right) \\ & - 8P\left((X_1, X_3) < (X_2, X_4), (Y_1, Y_4) < (Y_2, Y_3)\right).\end{aligned}$$

Bergsma-Dassios-Yanagimoto's correlation (continued)

Definition 4. (Bergsma-Dassios-Yanagimoto's correlation ; τ^* , τ_n^*)

It is unbiasedly estimated by the correlation coefficient

$$\begin{aligned}\tau_n^*(X, Y) = & \frac{1}{n(n-1)(n-2)(n-3)} \\ & \times \sum_{i_1 \neq \dots \neq i_4} \left\{ I\left((X_{i_1}, X_{i_3}) < (X_{i_2}, X_{i_4})\right) + I\left((X_{i_2}, X_{i_4}) < (X_{i_1}, X_{i_3})\right) \right. \\ & \left. - I\left((X_{i_1}, X_{i_4}) < (X_{i_2}, X_{i_3})\right) - I\left((X_{i_2}, X_{i_3}) < (X_{i_1}, X_{i_4})\right) \right\} \\ & \times \sum_{i_1 \neq \dots \neq i_4} \left\{ I\left((Y_{i_1}, Y_{i_3}) < (Y_{i_2}, Y_{i_4})\right) + I\left((Y_{i_2}, Y_{i_4}) < (Y_{i_1}, Y_{i_3})\right) \right. \\ & \left. - I\left((Y_{i_1}, Y_{i_4}) < (Y_{i_2}, Y_{i_3})\right) - I\left((Y_{i_2}, Y_{i_3}) < (Y_{i_1}, Y_{i_4})\right) \right\},\end{aligned}$$

which is a rank-based U -statistic of order 4.

Consistency

Proposition 1. (Consistency of estimators)

If the joint distribution of X and Y is continuous, we have

$$\xi_n \xrightarrow{P} \xi, \quad D_n \xrightarrow{P} D, \quad R_n \xrightarrow{P} R, \quad \tau_n^* \xrightarrow{P} \tau^*.$$

Proposition 2. (Consistency of correlation measures)

If the joint distribution is continuous, ξ , R , and τ^* are consistent, that is, they are zero-valued if and only if the pair (X, Y) is independent. Additionally, D is consistent when the joint distribution is absolutely continuous with respect to the Lebesgue measure.

Proposition 3. (Limiting null distributions)

Suppose that the joint distribution $F = F_X F_Y$ (X and Y are independent) is continuous. Then, as $n \rightarrow \infty$,

$$\xi_n = O_P(n^{-1/2}), \quad D_n = O_P(n^{-1}), \quad R_n = O_P(n^{-1}), \quad \tau_n^* = O_P(n^{-1}).$$

Local power analysis

Preliminaries

- Let Δ represent the **degree of dependence** between X and Y . If $\Delta = 0$, X and Y are independent, whereas X and Y become strongly dependent as $|\Delta| \rightarrow 1$.
- To investigate the powers of each correlation coefficient, the following sequence of hypothesis tests is considered:

$$H_0 : \Delta = 0 \text{ vs. } H_{1,n}(\Delta_0) : \Delta = \Delta_n, \quad \text{where } \Delta_n = n^{-1/2} \Delta_0. \quad (2)$$

- **As $n \rightarrow \infty$, it is expected that rejecting H_0 under $H_{1,n}$ becomes more difficult.**
- This study considers two classical and well-used families of alternatives: (i) rotation alternatives, and (ii) mixture alternatives.

Rotation alternatives

Assumption 1. (Rotation alternatives)

Let V, W be two real-valued **independent** random variables that have mean zero and are absolutely continuous with Lebesgue densities f_V and f_W , respectively. For $\Delta \in (-1, 1)$, consider

$$\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} 1 & \Delta \\ \Delta & 1 \end{pmatrix} \begin{pmatrix} V \\ W \end{pmatrix}.$$

Let $f_{X,Y}(x, y; \Delta)$ denote the density of (X, Y) . We assume that the following properties hold:

- (i) the distributions of (X, Y) have a **common support** for all $\Delta \in (-1, 1)$;
- (ii) the density f_k is **absolutely continuous with nonconstant logarithmic derivative** $\rho_k = f'_k/f_k$ for $k = V, W$;
- (iii) the Fisher information of (X, Y) relative to Δ at the point 0, denoted by $I_{X,Y}(0)$, is **strictly positive**, and $\mathbb{E}V^2, \mathbb{E}W^2 < \infty$ and $\mathbb{E}\rho_V(V^2), \mathbb{E}\rho_W(W^2) < \infty$.

- Assumption 1(ii) and (iii) imply that $\mathbb{E}\rho_V(V), \mathbb{E}\rho_W(W) = 0$ and $I_{X,Y}(0) < \infty$.
- Assumption 1 is required to deal with **the limit distribution of the likelihood ratio** between H_0 and $H_{1,n}$.

Assumption 2. (Mixture alternatives)

Let F_1 and F_2 be fixed univariate distributions that are absolutely continuous with Lebesgue density f_1 and f_2 , respectively. Let $F_0 = F_1 F_2$ be **the product distribution yielding independence**, and let $G = F_0$ be a fixed bivariate distribution which is absolutely continuous and such that X and Y are **dependent under G** . Let the density functions of F_0 and G , denoted by f_0 and g , respectively, be continuous and have **compact supports**. Then define the following alternative model for the distribution of (X, Y) :

$$F_{X,Y} = (1 - \Delta)F_0 + \Delta G,$$

with $0 \leq \Delta \leq 1$. We assume that

- (i) the distribution G is absolutely continuous with respect to F_0 and $s(x, y) = g(x, y)/f_0(x, y) - 1$ is continuous;
- (ii) the conditional expectation $\mathbb{E}(s(X, Y) | X) = 0$ almost surely for $(X, Y) \sim F_0$;
- (iii) the function s is **not additively separable**, i.e., there do not exist univariate functions h_1 and h_2 such that $s(x, y) = h_1(x) + h_2(y)$;
- (iv) the Fisher information $I_{X,Y}(0) > 0$.

- Assumption 2(i), (ii), and (iv) play an analogous role to Assumption 1.
- Assumption 2(iii) **ensures a complexity of g** .

Theorem 1. (Power analysis)

Suppose that the sequences of local alternatives considered are formed such that Assumption 1 or 2 holds when considering a family of rotation or mixture alternatives, respectively. Then concerning any sequence of alternatives, given in (2):

- (i) for either of the two types of alternatives and for any fixed constant $\Delta_0 > 0$,

$$\lim_{n \rightarrow \infty} P(T_{\alpha}^{\xi_n} = 1 \mid H_{1,n}(\Delta_0)) = \alpha;$$

- (ii) for any local alternative family and any number $\beta > 0$, there exists a sufficiently large constant $C_{\beta} > 0$, depending only on β , such that as long as $\Delta_0 > C_{\beta}$,

$$\lim_{n \rightarrow \infty} P(T_{\alpha}^{\mu_n} = 1 \mid H_{1,n}(\Delta_0)) \geq 1 - \beta,$$

where $\mu_n \in \{D_n, R_n, \tau_n^*\}$.

Le Cam's third lemma

Lemma 1. (Le Cam's third lemma)

Let P_n and Q_n be a sequence of probability measures and X_n be a sequence of random variables. If

$$\left(X_n \quad \log \frac{dQ_n}{dP_n}\right)^T \xrightarrow{P_n} N\left(\begin{pmatrix} \mu \\ -\frac{1}{2}\sigma^2 \end{pmatrix}, \begin{pmatrix} \Sigma & \tau \\ \tau^T & \sigma^2 \end{pmatrix}\right),$$

then $X_n \xrightarrow{Q_n} N(\mu + \tau, \Sigma)$.

- We write $X_n \xrightarrow{P_n} X$ if $\mathbb{E}_{P_n} f(X_n) \rightarrow \mathbb{E}_P f(X)$ for all bounded continuous functions f .
- In our framework, Q_n represents an alternative distribution.
- Intuitively, Le Cam's third lemma implies that **a slight adjustment occurs to the mean of a test statistic under an alternative distribution** when n is large.

Proof of Theorem 1(i)

It suffices to prove that $n^{1/2}\xi_n \xrightarrow[\Delta_n]{d} N(0, 2/5)$. The key of the proof is to show that

$$\text{Cov}(n^{1/2}\xi_n, \Lambda_n) \rightarrow 0,$$

under the null hypothesis, where $\Lambda_n = \sum_{i=1}^n \log \frac{f(X_i, Y_i; \Delta_n)}{f(X_i, Y_i; 0)}$. Note that X and Y are independent. Also, it holds that $Y_{(1)}, \dots, Y_{(n)}$ are still independent and identically distributed, where (i) is such that $X_{(1)} < \dots < X_{(n)}$.

We use the following fact:

$$n^{1/2}\xi_n = -3n^{-1/2} \sum_{i=1}^{n-1} \Xi_{(i)} + o_P(1),$$

where

$$\begin{aligned} \Xi_{(i)} = & |F_Y(Y_{(i+1)}) - F_Y(Y_{(i)})| + F_Y(Y_{(i+1)})\left(1 - F_Y(Y_{(i+1)})\right) \\ & + F_Y(Y_{(i)})\left(1 - F_Y(Y_{(i)})\right) - \frac{2}{3}. \end{aligned}$$

Proof of Theorem 1(i) (continued)

Considering that $F_Y(Y_{(i)}) \sim U(0, 1)$, one can show that

$$\mathbb{E}(\Xi_{(i)} \mid Y_{(i)}) = \mathbb{E}(\Xi_{(i)} \mid Y_{(i+1)}) \equiv c,$$

which gives $\text{Cov}\left(\Xi_{(i)}, \frac{f'(X_{(i)}, Y_{(i)}; 0)}{f(X_{(i)}, Y_{(i)}; 0)}\right) = \mathbb{E}\left(\frac{f'(X_{(i)}, Y_{(i)}; 0)}{f(X_{(i)}, Y_{(i)}; 0)} \mathbb{E}(\Xi_{(i)} \mid Y_{(i)})\right) = c \mathbb{E}\left(\frac{f'(X_{(i)}, Y_{(i)}; 0)}{f(X_{(i)}, Y_{(i)}; 0)}\right) = 0$. Similarly, $\text{Cov}\left(\Xi_{(i)}, \frac{f'(X_{(i+1)}, Y_{(i+1)}; 0)}{f(X_{(i+1)}, Y_{(i+1)}; 0)}\right) = 0$.

Using this property, one can show $\text{Cov}(n^{-1/2} \sum_{i=1}^{n-1} \Xi_{(i)}, \Lambda_n) \rightarrow 0$, which gives the desired result. □

Proposition 4. (Rate optimality)

Concerning either of the two local alternative families and any sequence of alternatives given in (2), as long as the corresponding Assumption 1 or 2 holds, we have that for any number $\beta > 0$ satisfying $\alpha + \beta < 1$ there exists a constant $c_\beta > 0$, depending only on β , such that

$$\inf_{T_\alpha \in \mathcal{T}_\alpha} P(T_\alpha = 0 \mid H_{1,n}(c_\beta)) \geq 1 - \alpha - \beta,$$

for all sufficiently large n . Here the infimum is taken over all size- α tests.

- Combined with Theorem 1 and Proposition 4, this result entails that the size- α test based on **Chatterjee's correlation coefficient is rate suboptimal, while the others achieve rate optimality.**
- The main reason for this result is that **Chatterjee's correlation coefficient cannot adjust its asymptotic behavior** under the alternative scenarios.

Simulation results

Simulation settings

(a) (X, Y) is given by the rotation alternative, where V and W are standard Gaussian and $\Delta_0 = 2$.

(b) (X, Y) is given by the mixture alternative, where

$$F_0(x, y) = U(x)U(y),$$

$$G(x, y) = U(x)U(y)\left(1 + (1 - U(x))(1 - U(y))\right),$$

where $U(\cdot)$ denoting the CDF of $U(-1, 1)$, and $\Delta_0 = 10$.

(c) (X, Y) is given by the mixture alternative, where the density functions of F_0 and G are

$$f_0(x, y) = u(x)u(y),$$

$$g(x, y) = u(x)u(y)\left(1 + |1 - 2U(x)|(1 - 2U(x))\right),$$

where $u(\cdot)$ denoting the density of $U(-1, 1)$, and $\Delta_0 = 20$.

Simulation settings

Generalized rotation alternatives

Let V, W be two real-valued **independent** random variables that have mean zero and are absolutely continuous with Lebesgue densities f_V and f_W , respectively. For $\Delta \in (-1, 1)$, consider

$$X = V, \quad Y = \Delta g(V) + W.$$

- (d) (X, Y) is given by the generalized rotation alternative, where $V \sim U(-1, 1)$, $W \sim N(0, 1)$, g takes values $-3, 2, -4, -3$ in the intervals $[-1, -0.5)$, $[-0.5, 0)$, $[0, 0.5)$, $[0.5, 1]$, respectively, and $\Delta_0 = 3$.
- (e) (X, Y) is given by the generalized rotation alternative, where $V \sim U(-1, 1)$, $W \sim N(0, 1)$, $g(t) = |t + 0.5|I(t < 0) + |t - 0.5|I(t \geq 0)$, and $\Delta_0 = 60$.
- (f) (X, Y) is given by the generalized rotation alternative, where $V \sim U(-1, 1)$, $W \sim N(0, 1)$, $g(t) = \cos(2\pi t)$, and $\Delta_0 = 12$.

Comparison of the computation time

Table 1. *Comparison of the computation time, measured as the total time in seconds of 1000 replicates, for all five correlation statistics*

n	ξ_n	ξ_n^*	D_n	R_n	τ_n^*
500	0.157	12.57	0.158	0.263	0.253
1000	0.239	33.75	0.267	0.505	0.468
5000	1.655	401.4	1.823	3.601	3.087
10000	3.089	1152.6	3.315	7.607	7.132

- ξ_n^* is another estimator for ξ based on kernel technique.
- It is known that ξ_n , D_n , R_n and τ_n^* can all be computed in $O(n \log n)$ time, while ξ_n^* necessitates a time complexity of $O(n^{5/3})$.

Comparison of the empirical powers

Table 2. Empirical powers, based on 1000 replicates, of the five competing tests in Example 4

n	ξ_n	ξ_n^*	D_n	R_n	τ_n^*	ξ_n	ξ_n^*	D_n	R_n	τ_n^*
Results for Example 4(a)						Results for Example 4(d)				
500	0.103	0.178	0.954	0.955	0.957	0.443	0.122	0.913	0.921	0.919
1000	0.067	0.106	0.956	0.956	0.956	0.285	0.111	0.923	0.928	0.927
5000	0.043	0.078	0.953	0.952	0.952	0.081	0.083	0.936	0.936	0.937
10000	0.045	0.058	0.951	0.952	0.952	0.081	0.052	0.955	0.954	0.955
Results for Example 4(b)						Results for Example 4(e)				
500	0.087	0.138	0.898	0.896	0.897	0.719	1.000	0.654	0.635	0.643
1000	0.067	0.089	0.900	0.900	0.899	0.486	1.000	0.700	0.682	0.692
5000	0.059	0.082	0.891	0.890	0.891	0.146	1.000	0.735	0.735	0.736
10000	0.052	0.045	0.911	0.914	0.915	0.105	0.997	0.754	0.752	0.752
Results for Example 4(c)						Results for Example 4(f)				
500	0.088	0.559	0.412	0.404	0.410	0.688	1.000	0.635	0.603	0.611
1000	0.066	0.408	0.390	0.391	0.396	0.459	1.000	0.669	0.655	0.660
5000	0.060	0.327	0.363	0.364	0.364	0.141	1.000	0.717	0.712	0.713
10000	0.048	0.248	0.392	0.395	0.396	0.100	0.994	0.726	0.730	0.728

Conclusion

Conclusion

Table 3. *Properties of the five rank correlation coefficients in Definitions 1–5; the bivariate distribution families are defined in (6) and (15)*

	μ_n	ξ_n	ξ_n^*	D_n	R_n	τ_n^*
(i) Computational efficiency	$F \in \mathcal{F}^c$ $F \in \mathcal{F}$	$O(n \log n)$ $O(n \log n)$	$O(n^{5/3})$ —	$O(n \log n)$ $O(n \log n)$	$O(n \log n)$ $O(n^2)$	$O(n \log n)$ $O(n^2)$
(ii) Consistency of correlation measures		$F \in \mathcal{F}^*$	$F \in \mathcal{F}^*$	$F \in \mathcal{F}^{ac}$	$F \in \mathcal{F}$	$F \in \mathcal{F}^{\tau^*}$
(ii') Consistency of independence tests		$F \in \mathcal{F}^c$	$F \in \mathcal{F}^{DSS}$	$F \in \mathcal{F}^{ac}$	$F \in \mathcal{F}^c$	$F \in \mathcal{F}^c$
(iii) Statistical efficiency	rotation mixture	rate-suboptimal rate-suboptimal	— —	rate-optimal rate-optimal	rate-optimal rate-optimal	rate-optimal rate-optimal

Reference

- Angus, J. E. (1995). A coupling proof of the asymptotic normality of the permutation oscillation. *Probability in the Engineering and Informational Sciences*, 9(4), 615-621.
- Chatterjee, S. (2021). A new coefficient of correlation. *Journal of the American Statistical Association*, 116(536), 2009-2022.
- Dette, H., Siburg, K. F., & Stoimenov, P. A. (2013). A Copula-Based Non-parametric Measure of Regression Dependence. *Scandinavian Journal of Statistics*, 40(1), 21-41.
- Shi, H., Drton, M., & Han, F. (2022). On the power of Chatterjee's rank correlation. *Biometrika*, 109(2), 317-333.